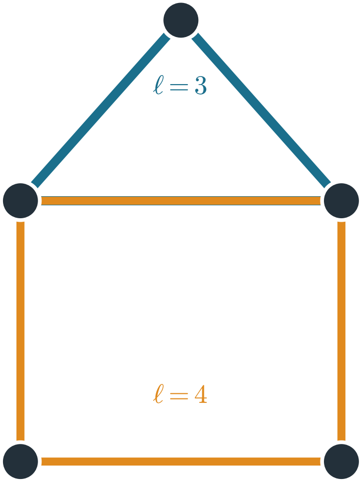
The Ihara zeta counts prime non-backtracking cycles; its reciprocal is $\det(I - uB)$

two prime cycles
(closed, non-backtracking, not a power)



$$\zeta_G(u) = \prod_{[C]\ \text{prime}} \frac{1}{1 - u^{\ell(C)}}$$

$$\zeta_G(u)^{-1} = \det(I - uB)$$

infinite Euler product = finite determinant

the object Bass's formula computes (this paper)